

Rubric

Problem 1. (10 points) Imagine human space flight from Earth to Mars in the following two stages.

- a. 5 pts total; -1 for not writing out the equation for Δt ; -1 for incorrect escape velocity; -1 for incorrect plugging in of numbers to get value for Δt . They should get at least 2 points for trying.

$$\Delta t = \frac{\Delta v}{a} = \frac{v_{\text{esc}}}{a} = \frac{11,171 \text{ m/s}}{3 \times 9.8 \text{ m/s}^2} = 380 \text{ s}$$

- b. 5 pts total; -1 for not writing out the equation for acceleration; -1 for incorrect plugging in of numbers. They should get at least 3 points for trying. $a = \frac{\Delta v}{\Delta t} = \frac{2900 \text{ m/s}}{90 \text{ s}} = 32.2 \text{ m/s}^2$

Problem 2. (10 points) Consider the film clip from The Martian shown in class which depicts Mark Watney's launch from the surface of Mars, starting from zero initial speed and accelerating to a final speed of 850 m/s. The change in speed takes place over a time interval of about 40 seconds.

- a. Convert the final speed of the spacecraft from meters per second to miles per hour using an online source (be sure to include a reference) or by converting miles to meters and hours to seconds.

$$850 \text{ m/s} = 1901.4 \text{ miles per hour (Wolfram Alpha)}$$

2 points; -1 for incorrect value; 1 pt for trying.

- b. Calculate the acceleration Watney experiences and compare it to the acceleration due to gravity on Earth.

$$a = \frac{\Delta v}{\Delta t} = \frac{850 \text{ m/s}}{40 \text{ s}} = 21.25 \text{ m/s}^2 \quad a/g = 21.25/9.8 = 2.17$$

4 points; -1 for not writing equation; -1 for incorrect value; -1 for no comparison

- c. 4 points; -1 for incorrect logic

Students should make an argument about whether they think it is realistic for him to have passed out. As long as it is reasonable, they can get points. (most, if not all, should get points)